

Convexity

Design and Analysis of Algorithms

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Definitions

- Convex sets
- Convex functions
 - In one dimension, $f''(x) \geq 0$
 - In multiple dimensions, $\nabla^2 f(x) \geq 0$ (positive semi-definite)
(“strongly” convex if $>$ instead of $\geq$)

Definitions

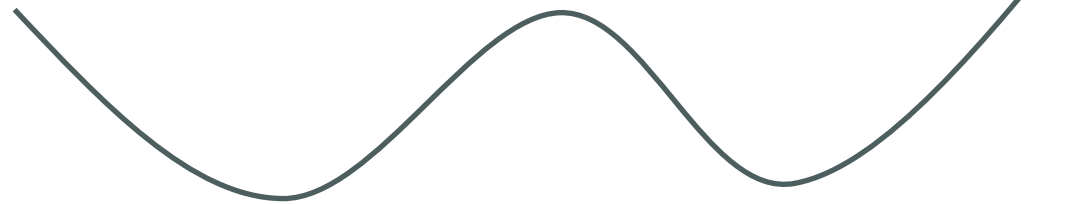
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- Convex functions
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(“strongly” convex if $>$ instead of $\geq$)
- Condition number $\kappa = M / m$, with $f''(x)$ in the range $[m, M]$
(in multiple dimensions, eigenvalues of $\nabla^2 f(x)$ live in $[m, M]$)
 - Only defined if strongly convex (otherwise $m = 0$).
 - Optimization algorithms usually much faster the closer κ is to 1.

Algorithmic Implications

- For a convex function, any local minimum is also a global minimum.
- In constrained optimization, things are much easier if we optimize over a convex set.
- In general, optimizing a convex function over a convex set is highly tractable; otherwise we are in more challenging heuristic territory.



Convex



Non-Convex